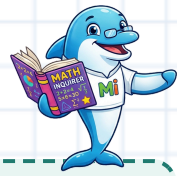
PICTURE EQUATIONS!

Name _____ Date _____

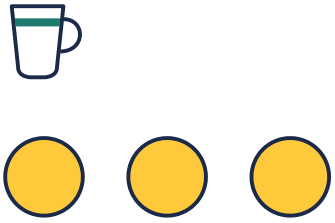


After question 4 you may choose: carry on, or try **Think harder**.

Each shape is a number. Use counters. Check your answer in the line.

No.S1

A cup hides some counters. 3 counters are outside the cup. There are 11 counters in all. Fill in the blank.



The cup holds ☐ counters.

No.S2

Each shape is a number. Find the number for the circle.

● + ■ = 10

■ = 4 ● = ☐

No.S3

Each shape is a number. Find the number for the heart.

♥ + ♥ = 16

♥ = ☐

No.S4

Each shape is a number. Find the number for the star.

★ + ★ + ★ = 15

★ = ☐

Name _____ Date _____

The same shape is always the same number. In our puzzles, different shapes are different numbers.

No.S5

Each shape is a number. The square is 4. Fill in the blanks.



(a) The 2 triangles make ☐☐.

(b) Each triangle is ☐.

No.S6

Ben says the circle is 7. Look at the picture.

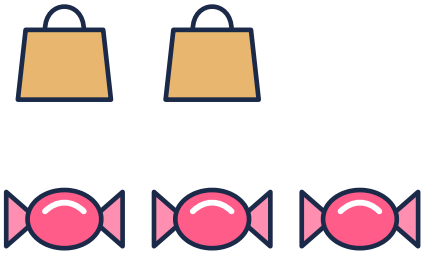


(a) Is Ben correct? Circle Yes or No. **Yes** **No**

(b) The circle is ☐.

No.S7

Each bag has the same number of sweets. 3 sweets are outside the bags. There are 13 sweets in all. Fill in the blanks.



(a) The 2 bags have ☐☐ sweets in all.

(b) Each bag has ☐ sweets.

REASONING ABOUT EQUALITY: PICTURE EQUATIONS

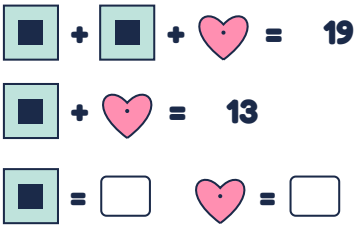
Think harder

Name _____ Date _____

Use counters or draw. Check your answer in every line. × means groups of. Choose any question.

No.T1

Each shape is a number. Find the number for each shape. Show your working.



No.T2

Kumar says the heart is 8 and the star is 7. Look at the picture. Decide if each statement is True or False.

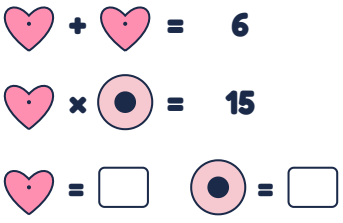


Statement	True	False
Line 1 is true with Kumar's numbers.	☐	☐
Line 2 is true with Kumar's numbers.	☐	☐

Put a tick (✓) in the correct box.

No.T3

Each shape is a number. Find the number for each shape.

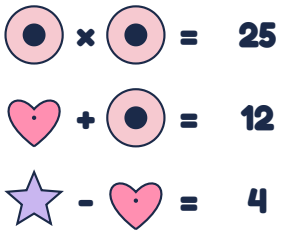


Name _____ Date _____

× means groups of. Choose any question. You do not have to finish them all.

No.G1

Each shape is a number. Find the number for each shape.



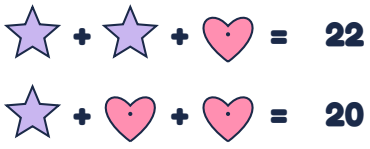
(a) The circle is ☐.

(b) The heart is ☐.

(c) The star is ☐.

No.G2

Each shape is a number. Fill in the blanks.



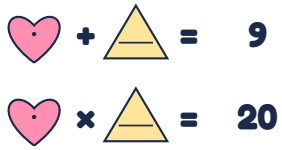
(a) One star and one heart make ☐☐.

(b) The star is ☐.

(c) The heart is ☐.

No.G3

Each shape is a number. The heart is greater than the triangle. Find the number for each shape.



(a) The heart is ☐.

(b) The triangle is ☐.

Name _____ Date _____

Investigation: One line or two?

First: The circle + the triangle = 10. One pair: circle 2, triangle 8. Write 2 more.

Next: The circle + the circle + the triangle = 14. Do any pairs fit both lines? Why?

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Each shape is a number. The triangle + the triangle = 18. Find the number for the triangle.

The triangle is ☐.

No.E2

The heart + the square = 13. The square is 5. Find the number for the heart. Choose the correct answer. Write its number in the bracket.

(1) 5

(2) 6

(3) 7

(4) 8

( )

No.E3

Each shape is a number. The star is 5. Find the number for each circle.



Each circle is ☐.

Look-back

There are no right answers. You can pass. Nobody reads it out.

Something I did well: .....

The hardest part was ... ☐ finding the number for one shape ☐ using two lines ☐ checking in every line ☐ something else

What helped me: ☐ counters and shape cards ☐ drawing it ☐ checking my answer ☐ I was not stuck

Next time I will try: ☐ using counters ☐ checking my answer ☐ drawing it ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

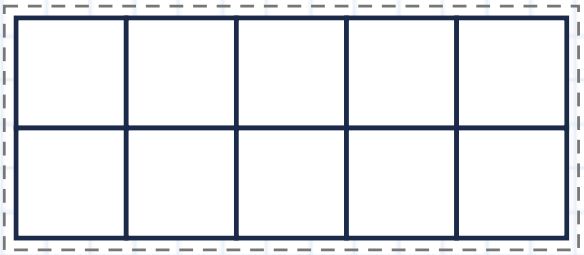
Name _____ Date _____

Kit page: ten-frames and counters

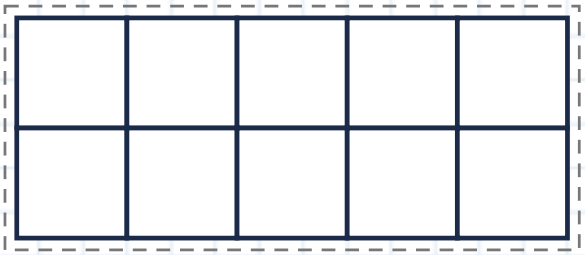
A grown-up cuts out the 20 counters (10 red, 10 yellow) before the lesson. Red counters can hide in a cup; yellow counters can be the loose ones. The ten-frames stay on the page.

Print this page on its own sheet, or on one side only, so cutting never spoils another page. A grown-up cuts; children do not use scissors. Keep the pieces in an envelope with your name. We use them again.

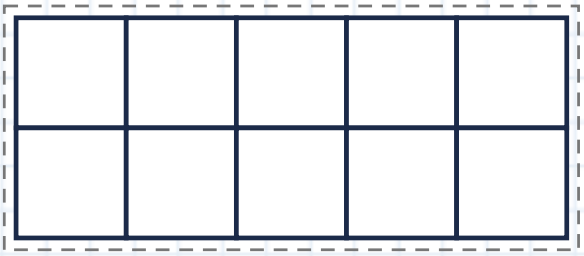
Ten-frame 1



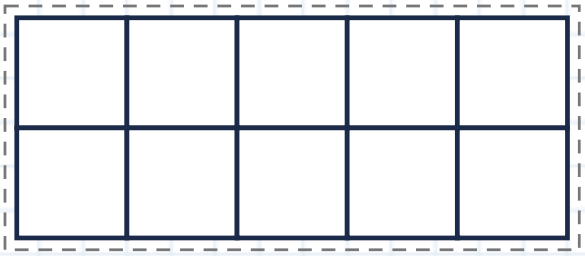
Ten-frame 2



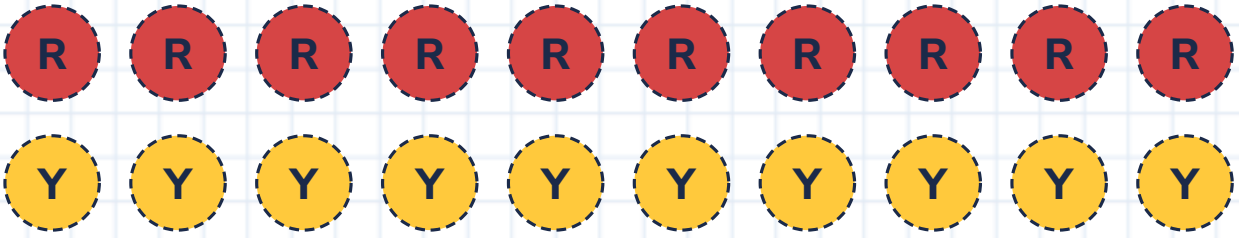
Ten-frame 3



Ten-frame 4



Counters: cut out all 20 (10 red marked R, 10 yellow marked Y)



Name _____ Date _____

Kit page: shape cards

A grown-up cuts out the 15 shape cards before the lesson: three each of circle, square, triangle, heart and star. Each card is big enough for 6 counters. Children may draw the shape on its card. Keep the cards in an envelope.

Print this page on its own sheet, or on one side only, so cutting never spoils another page. A grown-up cuts; children do not use scissors. Keep the pieces in an envelope with your name. We use them again.

circle	circle	circle
square	square	square
triangle	triangle	triangle
heart	heart	heart
star	star	star